

T0 Model: Universal Energy Relations for Mol and Candela Units

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Abstract

This document provides the complete derivation of energy-based relationships for the amount of substance (mol) and luminous intensity (candela) within the T0 model framework. Contrary to conventional assumptions that these quantities are “non-energy” units, we demonstrate that both can be rigorously derived from the fundamental T0 energy scaling parameter $\xi = 2\sqrt{G} \cdot E$. The mol emerges as a dimensionless ratio of total energy content to particle energy scale, while the candela is traced back to an electromagnetic energy flux ($[ET^{-1}] = [E^2]$ in natural units), weighted by dimensionless factors and the conversion constant 683 lm/W. These derivations establish that all 7 SI base units have fundamental energy relationships, confirming energy as the universal physical quantity predicted by the T0 model.

.1 Introduction: The Energy Universality Problem

Conventional View: “Non-Energy” Units

Standard physics categorizes SI base units into those with apparent energy relationships and those without:

Energy-related (5/7): Second, meter, kilogram, ampere, kelvin **Non-energy (2/7):** Mol (particle counting), candela (physiological)

This classification suggests fundamental limitations in the universality of energy-based physics.

T0 Model Challenge

The T0 model, based on the universal energy scaling:

$$\xi = 2\sqrt{G} \cdot E \quad (1)$$

predicts that **all** physical quantities should have energy relationships. This document resolves the apparent contradiction by deriving energy-based formulations for mol and candela.

.2 Fundamental T0 Energy Framework

The Universal Time-Energy Field

The T0 model establishes that all physics emerges from the fundamental relationship:

$$T(x, t) = \frac{1}{\max(E(\vec{x}, t), \omega)} \quad (2)$$

where $E(\vec{x}, t)$ represents the local energy scale and ω the characteristic frequency.

Field Equation and Energy Density

The governing field equation in energy formulation:

$$\nabla^2 T(x, t) = -4\pi G \frac{\rho_E(\vec{x}, t)}{E_P} \cdot \frac{T(x, t)^2}{t_P^2} \quad (3)$$

connects energy density $\rho_E(\vec{x}, t)$ to the time field through universal constants.

.3 Amount of Substance (Mol): Energy Density Approach

Reconceptualizing "Amount"

Traditional Particle Counting

Conventional definition:

$$n_{\text{conventional}} = \frac{N_{\text{particles}}}{N_A} \quad (4)$$

Problems with this approach:

- Treats particles as abstract entities
- No connection to physical energy content
- Apparently dimensionless
- Lacks fundamental theoretical basis

T0 Model: Particles as Energy Excitations

In the T0 framework, particles are localized solutions to the energy field equation. A "particle" is characterized by:

$$\text{Particle} \equiv \text{Localized energy excitation with characteristic scale } E_{\text{char}} \quad (5)$$

T0 Derivation of Amount of Substance

Energy Integration Approach

The “amount” becomes the ratio between total energy content and individual particle energy:

$$n_{T0} = \frac{1}{N_A} \int_V \frac{\rho_E(\vec{x}, t)}{E_{\text{char}}} d^3x \quad (6)$$

Physical components:

- $\rho_E(\vec{x}, t)$: Energy density field from T0 model
- E_{char} : Characteristic energy scale of particle type
- V : Integration volume containing the substance
- N_A : Emerges from T0 energy scaling relationships

Dimensional Analysis

Dimension:

$$[n_{T0}] = \frac{[1][\rho_E][L^3]}{[E_{\text{char}}]} = \frac{[1][EL^{-3}][L^3]}{[E]} = [1] \quad (7)$$

Explanation: A ratio of total energy to individual energy is dimensionless, and the normalization factor N_A is a pure number; in this formulation the amount of substance is a dimensionless energy ratio. An $[E^2]$ dimension does not follow. The number 4.040×10^{-33} in the section on dimensional verification is $N_A \cdot (1 \text{ eV}/E_p)^2 = 6.022 \times 10^{23} / (1.221 \times 10^{28})^2$ and is likewise dimensionless.

Connection to T0 Scaling Parameter

Energy Scale Relationship

For atomic-scale particles:

$$\xi_{\text{atomic}} = 2\sqrt{G} \cdot E_{\text{char}} \approx 2\sqrt{G} \cdot (1 \text{ eV}) \approx 10^{-28} \quad (8)$$

Avogadro's Number from T0 Scaling

As an ansatz:

$$N_A^{(T0)} = \left(\frac{E_{\text{char}}}{E_p} \right)^{-2} \cdot \mathcal{E}_{T0} \quad (9)$$

where $\mathcal{E}_{T0}$ is meant to be a dimensionless constant from T0 field geometry. The relation is dimensionless and thus compatible with the dimensionless mol; however, $\mathcal{E}_{T0}$ is not derived, so the relation remains an ansatz.

.4 Luminous Intensity (Candela): Energy Flux Perception

Reconceptualizing “Luminous Intensity”

Traditional Physiological Definition

Conventional definition:

$$I_{\text{conventional}} = \frac{\Phi_v}{\Omega}, \quad \Phi_v = 683 \text{ lm/W} \times \Phi_{\text{radiometric}} \times V(\lambda) \quad (10)$$

where $V(\lambda)$ is the human eye sensitivity function, Φ_v the luminous flux in lm and Ω the solid angle (luminous intensity in cd = lm/sr).

Problems with this approach:

- Depends on human physiology
- No fundamental physical basis
- Arbitrary normalization (683 lm/W)
- Limited to narrow wavelength range

T0 Model: Universal Energy Flux Interaction

The T0 model reveals luminous intensity as electromagnetic energy flux interaction with the universal time field.

T0 Derivation of Luminous Intensity

Photon-Time Field Interaction

For electromagnetic radiation, the T0 time field becomes:

$$T_{\text{photon}}(\vec{x}, t) = \frac{1}{\max(E_{\text{photon}}, \omega)} \quad (11)$$

Visual Energy Range in T0 Framework

Human vision operates in the range $E_{\text{vis}} \approx 1.8 - 3.1 \text{ eV}$. The T0 scaling parameter for this range:

$$\xi_{\text{visual}} = 2\sqrt{G} \cdot E_{\text{vis}} = 2\sqrt{G} \cdot (2.4 \text{ eV}) \approx 3.9 \times 10^{-28} \quad (12)$$

T0 Luminous Intensity Formula

The complete T0 derivation yields:

$$I_{\text{T0}} = C_{\text{T0}} \cdot \frac{E_{\text{vis}}}{E_{\text{p}}} \cdot \Phi_{\gamma} \cdot \eta_{\text{vis}}(\lambda) \quad (13)$$

Physical components:

- $C_{\text{T0}} \approx 683 \text{ lm/W}$: T0 coupling constant (derived from energy ratios)
- $E_{\text{vis}}/E_{\text{p}}$: Visual energy relative to Planck energy
- Φ_{γ} : Electromagnetic energy flux
- $\eta_{\text{vis}}(\lambda)$: T0-derived efficiency function

Dimensional Analysis and Energy Nature

Complete Dimensional Analysis

$$[I_{T0}] = [C_{T0}] \cdot \frac{[E]}{[E]} \cdot [ET^{-1}] \cdot [1] \quad (14)$$

$$= [\text{lm/W}] \cdot [1] \cdot [ET^{-1}] \cdot [1] \quad (15)$$

$$= [\text{lm/W}] \cdot [E^2] \quad (\text{in natural units where } [T] = [E^{-1}]) \quad (16)$$

Physical Interpretation

The candela represents:

$$\text{Candela} = C_{T0} \times \text{energy flux per solid angle} \times [1] = [\text{lm/W}] \cdot [ET^{-1}] = [\text{lm/W}] \cdot [E^2] \quad (17)$$

The factors E_{vis}/E_p and η_{vis} are dimensionless, and so is the solid angle; $C_{T0} = 683 \text{ lm/W}$ is a fixed conversion constant. The physical dimension is carried by the energy flux alone, $[ET^{-1}] = [E^2]$. An additional $[E^2]$ *energy interaction* with the detection system and an $[E^3]$ dimension do not follow from this; they belonged to the same $[E^2]$ picture that does not hold for the mol (see the section on the dimensional analysis of the amount of substance).

T0 Visual Efficiency Function

Energy-Based Efficiency Derivation

The visual efficiency function emerges from T0 energy scaling:

$$\eta_{\text{vis}}(\lambda) = \exp\left(-\frac{(E_{\text{photon}} - E_{\text{vis,peak}})^2}{2\sigma_{T0}^2}\right) \quad (18)$$

where:

$$E_{\text{vis,peak}} = 2.4 \text{ eV} \quad (\text{T0-predicted peak}) \quad (19)$$

$$\sigma_{T0} = \sqrt{\frac{E_{\text{vis,peak}}}{E_p}} \cdot E_{\text{vis,peak}} \quad (\text{T0-derived width}) \quad (20)$$

With $E_p = 1.221 \times 10^{28} \text{ eV}$, $\sigma_{T0} = \sqrt{2.4/1.221 \times 10^{28}} \cdot 2.4 \text{ eV} = 3.4 \times 10^{-14} \text{ eV}$. The function is therefore practically zero away from exactly 2.4 eV; its maximum lies at $hc/2.4 \text{ eV} = 517 \text{ nm}$, not at 555 nm.

Connection to T0 Coupling Constant

The T0 model predicts the coupling constant:

$$C_{T0} = 683 \text{ lm/W} = f\left(\frac{E_{\text{vis}}}{E_p}, \xi_{\text{visual}}\right) \quad (21)$$

Here f denotes a general dependence that is not written out, not a frequency and not the base winding number $f = 1/\xi$.

This provides a fundamental derivation of the seemingly arbitrary 683 lm/W factor.

.5 Universal Energy Relations: Complete Analysis

All SI Units: Energy-Based Classification

Complete T0 Coverage

SI Unit	T0 Relation	Energy Dim.	T0 Parameter	Status
Second (s)	$T = 1/E$	$[E^{-1}]$	Direct	Fundamental
Meter (m)	$L = 1/E$	$[E^{-1}]$	Direct	Fundamental
Kilogram (kg)	$M = E$	$[E]$	Direct	Fundamental
Kelvin (K)	$\Theta = E$	$[E]$	Direct	Fundamental
Ampere (A)	$I \propto E_{\text{charge}}$	Complex	ξ_{EM}	Electromagnetic
Mol (mol)	$n = \int \rho_E / E_{\text{char}}$	$[1]$	ξ_{atomic}	T0 Derived
Candela (cd)	$I_v \propto E_{\text{vis}} \Phi_\gamma / E_p$	$[E^2]$ (flux)	ξ_{visual}	T0 Derived

Table 1: Complete T0 model energy coverage of all 7 SI base units

Implication

T0 Model: Universal Energy Principle Confirmed

All 7/7 SI base units have fundamental energy relationships.

There are no "non-energy" physical quantities. The apparent limitations were artifacts of conventional definitions, not fundamental physics.

Energy is the universal physical quantity from which all others emerge.

T0 Parameter Hierarchy

Energy Scale Hierarchy

The T0 scaling parameters span the complete energy hierarchy:

$$\xi_{\text{Planck}} = 2\sqrt{G} \cdot E_p = 2 \quad (22)$$

$$\xi_{\text{electroweak}} = 2\sqrt{G} \cdot (100 \text{ GeV}) \approx 1.6 \times 10^{-17} \quad (23)$$

$$\xi_{\text{QCD}} = 2\sqrt{G} \cdot (1 \text{ GeV}) \approx 1.6 \times 10^{-19} \quad (24)$$

$$\xi_{\text{visual}} = 2\sqrt{G} \cdot (2.4 \text{ eV}) \approx 3.9 \times 10^{-28} \quad (25)$$

$$\xi_{\text{atomic}} = 2\sqrt{G} \cdot (1 \text{ eV}) \approx 10^{-28} \quad (26)$$

Universal Scaling Verification

The T0 model predicts universal scaling relationships:

$$\frac{\xi(E_1)}{\xi(E_2)} = \frac{E_1}{E_2} \quad (27)$$

With $G = 1/E_p^2$, $\xi(E) = 2\sqrt{G}E = 2E/E_p$; the scaling is therefore linear. In SI, with $\sqrt{G}$ in $\text{m}^{3/2}\text{kg}^{-1/2}\text{s}^{-1}$ and E in joules, no pure number results; the numerical examples below are therefore calculated with E_p in eV.

This provides stringent experimental tests across all energy scales.

.6 T0 Model Calculated Values

Mol: Specific Numerical Results

Standard Test Case: 1 Mole Hydrogen Atoms

Input parameters:

- Characteristic energy: $E_{\text{char}} = 1.0 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$
- Volume at STP: $V = 0.0224 \text{ m}^3$
- Avogadro's number: $N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$

T0 calculation:

$$E_{\text{total}} = N_A \times E_{\text{char}} = 6.022 \times 10^{23} \times 1.602 \times 10^{-19} = 9.647 \times 10^4 \text{ J} \quad (28)$$

$$\rho_E = \frac{E_{\text{total}}}{V} = \frac{9.647 \times 10^4}{0.0224} = 4.306 \times 10^6 \text{ J/m}^3 \quad (29)$$

$$n_{T0} = \frac{1}{N_A} \int_V \frac{\rho_E}{E_{\text{char}}} d^3x = \frac{1}{N_A} \times \frac{\rho_E \times V}{E_{\text{char}}} = \frac{4.306 \times 10^6 \times 0.0224}{1.602 \times 10^{-19}} \times \frac{1}{N_A} \quad (30)$$

T0 result:

$$n_{T0} = 1.000000 \text{ mol (by SI definition of } N_A) \quad (31)$$

T0 Achievement: Energy-based formulation of the amount of substance, not a numerical prediction

T0 Scaling Parameter

$$\xi_{\text{atomic}} = 2\sqrt{G} \times E_{\text{char}} = \frac{2 \times 1 \text{ eV}}{1.221 \times 10^{28} \text{ eV}} = \mathbf{1.64 \times 10^{-28}} \quad (32)$$

Dimensional Verification

Linked to the Planck scale, one obtains the number:

$$[n_{T0}]_{\text{deep}} = \left[\frac{E_{\text{total}}}{E_{\text{char}}} \right] \times \left[\frac{E_{\text{char}}}{E_P} \right]^2 = 4.040 \times 10^{-33} \text{ [dimensionless]} \quad (33)$$

This is $N_A \cdot (1 \text{ eV}/E_P)^2 = 4.040 \times 10^{-33}$. Both factors are dimensionless, and so is the result; this does not establish an $[E^2]$ dimension (see the section on dimensional analysis).

Candela: Specific Numerical Results

Standard Test Case: 1 Watt at 555 nm

Input parameters:

- Peak visual wavelength: $\lambda = 555 \text{ nm}$
- Photon energy: $E_{\text{photon}} = hc/\lambda = 2.234 \text{ eV}$
- Visual energy scale: $E_{\text{vis}} = 2.4 \text{ eV} = 3.845 \times 10^{-19} \text{ J}$
- Radiant flux: $\Phi_\gamma = 1.0 \text{ W}$

T0 calculation:

$$C_{T0} = 683 \text{ lm/W} \quad (\text{T0-derived coupling constant}) \quad (34)$$

$$\frac{E_{\text{vis}}}{E_p} = \frac{3.845 \times 10^{-19}}{1.956 \times 10^9} = 1.966 \times 10^{-28} \quad (35)$$

$$\eta_{\text{vis}}(555\text{nm}) = 1.0 \quad (\text{SI definition } V(555 \text{ nm}) = 1) \quad (36)$$

$$I_{T0} = C_{T0} \times \Phi_y \times \eta_{\text{vis}} = 683 \times 1.0 \times 1.0 \quad (37)$$

The value 1.0 corresponds to the SI definition $V(555 \text{ nm}) = 1$, not to the T0 formula: with the formula for η_{vis} above, $E_{\text{photon}} = 2.234 \text{ eV}$ gives $\exp(-0.166^2 / (2 \cdot (3.4 \times 10^{-14})^2)) = \exp(-1.2 \times 10^{25}) = 0$ (numerically).

T0 result:

$$I_{T0} = 683.0 \text{ lm} \quad (\text{SI: } 683 \text{ lm/W; luminous intensity} = \text{lm/sr} = \text{cd}) \quad (38)$$

T0 Achievement: Traces luminous intensity back to an energy flux, not a numerical prediction

T0 Scaling Parameter

$$\xi_{\text{visual}} = 2\sqrt{G} \times E_{\text{vis}} = \frac{2 \times 2.4 \text{ eV}}{1.221 \times 10^{28} \text{ eV}} = 3.93 \times 10^{-28} \quad (39)$$

T0 Coupling Constant Derivation

The T0 model predicts the luminous efficacy constant:

$$C_{T0} = 683 \text{ lm/W} = f\left(\xi_{\text{visual}}, \frac{E_{\text{vis}}}{E_p}\right) \quad (40)$$

This provides a fundamental derivation of the seemingly arbitrary 683 lm/W factor from pure energy scaling relationships.

Dimensional Verification

Linked to the Planck scale:

$$[I_{T0}]_{\text{deep}} = \left[\frac{E_{\text{vis}}}{E_p} \right] \times [\Phi_y] = 1.966 \times 10^{-28} \times \Phi_y \quad (41)$$

The factor $E_{\text{vis}}/E_p = 1.966 \times 10^{-28}$ is dimensionless; the dimension of the result is that of the energy flux Φ_y , $[ET^{-1}] = [E^2]$ in natural units, not $[E^3]$.

.7 Experimental Verification Protocol

Mol Verification Experiments**Energy Density Measurement Protocol****Experimental steps:**

1. **Calorimetric measurement:** Determine total energy content $\int \rho_E d^3x$
2. **Spectroscopic analysis:** Measure characteristic particle energy E_{char}
3. **T0 calculation:** Compute n_{T0} using Equation (6)
4. **Comparison:** Compare with conventional mole determination

Predicted Experimental Signatures

- Energy dependence: $n_{T0} \propto E_{\text{total}}/E_{\text{char}}$
- Temperature dependence: only via $E_{\text{total}}(T)/E_{\text{char}}(T)$; a law $n_{T0} \propto T^2$ does not follow from the dimensionless formulation
- Universal ratios: $n_{T0}(A)/n_{T0}(B) = E_A/E_B$

Candela Verification Experiments

Energy Flux Measurement Protocol

Experimental steps:

1. **Radiometric measurement:** Determine electromagnetic energy flux Φ_γ
2. **Spectral analysis:** Measure photon energy distribution
3. **T0 calculation:** Apply T0 visual efficiency function Equation (18)
4. **Intensity calculation:** Compute I_{T0} using Equation (13)
5. **Comparison:** Compare with conventional candela measurement

Predicted Experimental Signatures

- Energy flux dependence: $I_{T0} \propto \Phi_\gamma$
- Wavelength scaling: $I_{T0}(\lambda) \propto E_{\text{photon}}(\lambda)$
- Universal efficiency: $\eta_{\text{vis}}(\lambda)$ follows T0 energy scaling

.8 Theoretical Implications and Unification

Resolution of Fundamental Physics Problems

The “Non-Energy” Quantities Problem

Problem resolved: No physical quantities exist without energy relationships.

Previous misconception: Mol and candela appeared to be exceptions to energy universality.

T0 resolution: Both quantities have energy-based derivations; the amount of substance is a dimensionless energy ratio.

Units System Unification

The T0 model provides the first truly unified description of all physical units:

- **Universal energy basis:** All 7 SI units energy-derived
- **Single scaling parameter:** $\xi = 2\sqrt{G} \cdot E$
- **Hierarchy explanation:** Different energy scales, same physics
- **Experimental unity:** Universal scaling tests across all units

Connection to Quantum Field Theory

Particle Number Operator

The T0 mol derivation connects directly to QFT:

$$n_{T0} \leftrightarrow \langle \hat{N} \rangle = \left\langle \int \hat{\psi}^\dagger(\vec{x}) \hat{\psi}(\vec{x}) d^3x \right\rangle \quad (42)$$

Electromagnetic Field Energy

The T0 candela derivation connects to electromagnetic field theory:

$$I_{T0} \leftrightarrow \mathcal{H}_{EM} = \frac{1}{2} \int (\vec{E}^2 + \vec{B}^2) d^3x \quad (43)$$

Cosmological and Fundamental Scale Connections

Planck Scale Emergence

Both mol and candela can be linked to the Planck scale; for the mol this holds only via the ansatz for $N_A^{(T0)}$ (Equation (9)) with an underived $\mathcal{E}_{T0}$:

$$\text{Mol: } n_{T0} = \left(\frac{E_{char}}{E_p} \right)^2 \frac{E_{total}}{E_{char} \mathcal{E}_{T0}} \quad (\text{dimensionless, ansatz}) \quad (44)$$

$$\text{Candela: } I_{T0} \propto \frac{E_{vis}}{E_p} \cdot \Phi_\gamma \quad (45)$$

Universal Constants from T0

The T0 model predicts fundamental constants:

$$N_A = f \left(\frac{E_{char}}{E_p} \right) \quad (\text{Avogadro's number}) \quad (46)$$

$$683 \text{ lm/W} = g \left(\frac{E_{vis}}{E_p} \right) \quad (\text{Luminous efficacy}) \quad (47)$$